USB POWER / ALL-RAIL PROTECTION AND RESET

S6 schematic only. All three converters retained: main and + rails TPS16630: negative LM5067 + MOSFET.

USB PD and main eFuse

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A nominal

File: 01_usb_input.kicad_sch

+12 V converter and output eFuse

MAIN_OK
P12_OK
RUN
V3V3
VIN
+12V
P12_RAW

TPS62933F -> TPS16630
+12 V protected output

File: 02_positive_12v.kicad_sch

+5 V converter and output eFuse

MAIN_OK
P5_OK
RUN
V3V3
VIN
+5V

TPS62933F -> TPS16630
+5 V protected output

File: 03_positive_5v.kicad_sch

Negative converter - raw rail

P12_RAW
VIN
N12_RAW

LMR38020F inverting buck-boost
RAW negative supply

File: 04_negative_converter.kicad_sch

Negative rail protection and level shifting

N12_RAW
NOT_RUN
V3V3
VIN
−12V
N12_STATUS

LM5067 + external N-MOSFET
1.0 A nominal / optical control

File: 05_negative_protection.kicad_sch

Protected outputs - voltage monitors and panel LEDs

+12V
+5V
−12V
MAIN_LED_OK
N12_STATUS
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK

Parallel Eurorack headers
MAIN, +12, +5 and −12 LEDs

File: 07_outputs_status.kicad_sch

Always-on control - all-rail fault latch and RESET

MAIN_OK
N12_OK
NOT_RUN
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always-on LDO / retained ARMED + FAULT
Manual RESET disables all three rails

File: 06_fault_reset.kicad_sch

FAULT POLICY
Each rail detects independently.
Persistent fault -> all OFF until RESET.
Main LED is not merely a +5 V lamp.

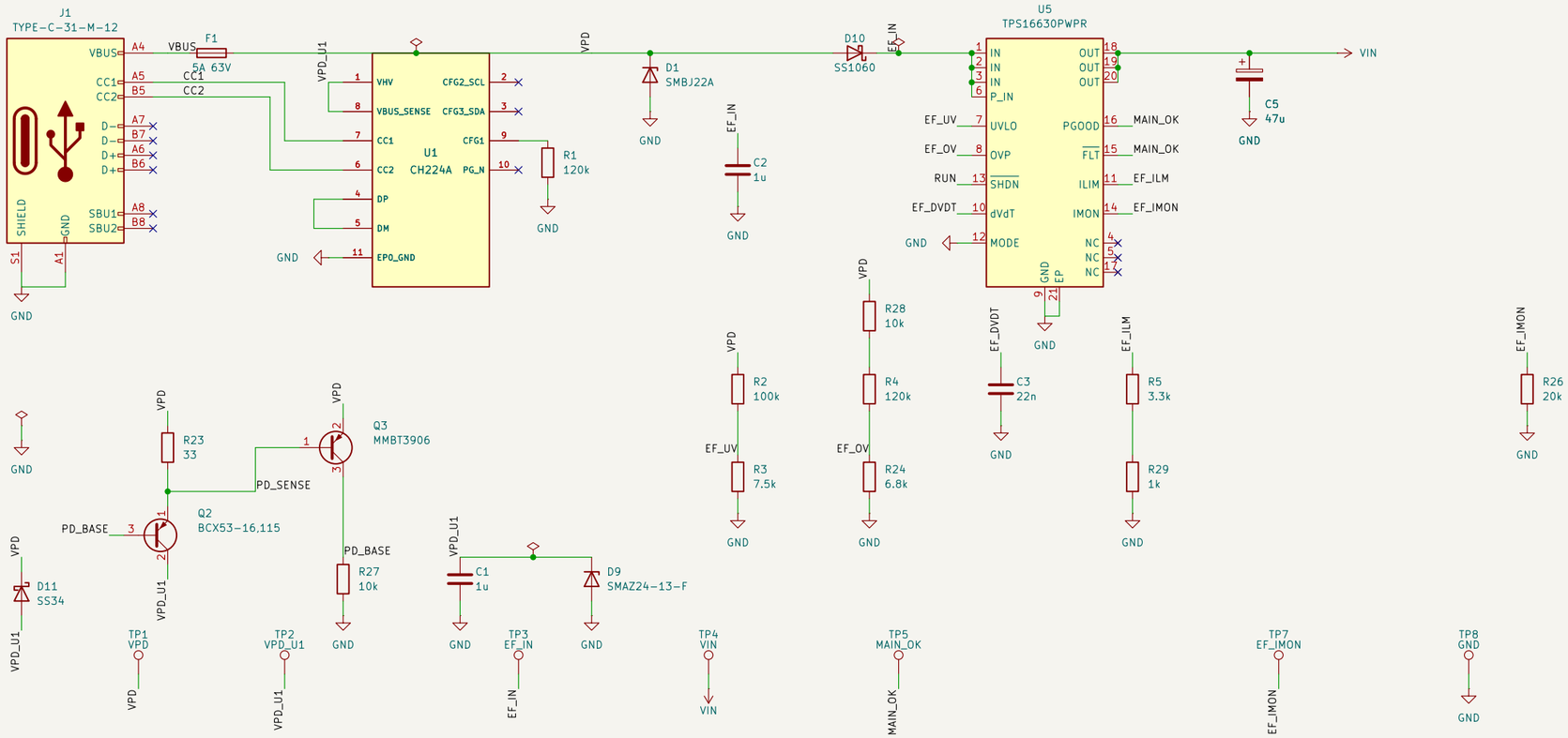
OUTPUTS: +12 / +5 / −12
Nominal limits: 2.647 / 2.647 / 1.0 A
Tolerance and thermal limits apply.
Not a guaranteed 54.5 W simultaneous rating.

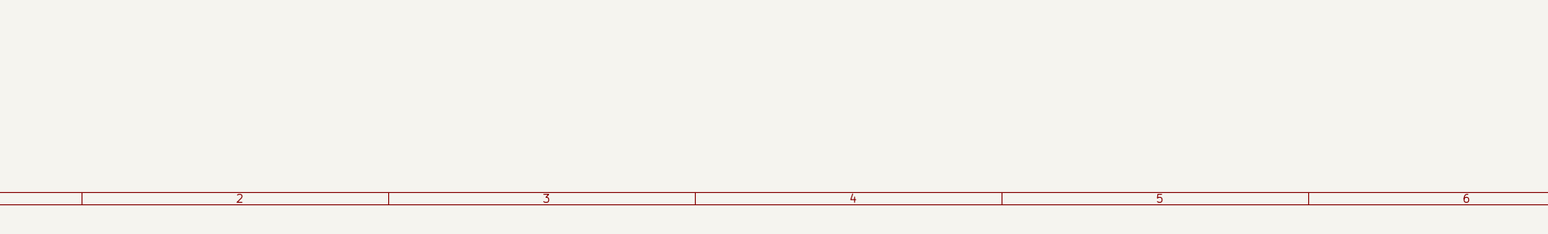
Design notes and budget: docs/design.md
S6: per-rail protection; all-rail fault latch/reset; four status LEDs
Big Modular
Sheet: /
File: usb-power-45w.kicad_sch

Title: USB power - wired power flow and protection overview

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 1/8

USB PD AND MAIN EFUSE





20 V PD request: U5 TPS16630 remains 60 V, 4.186 A nominal. RUN from retained fault latch.
MAIN_OK is wired FLT + PGOOD at 3.3 V, NOT a raw-voltage logic output.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇— MAIN_OK

RUN ◇— RUN

VIN ◇— VIN

VPD ◇— VPD

Schematic only — see design.md and s6-rail-protection.md

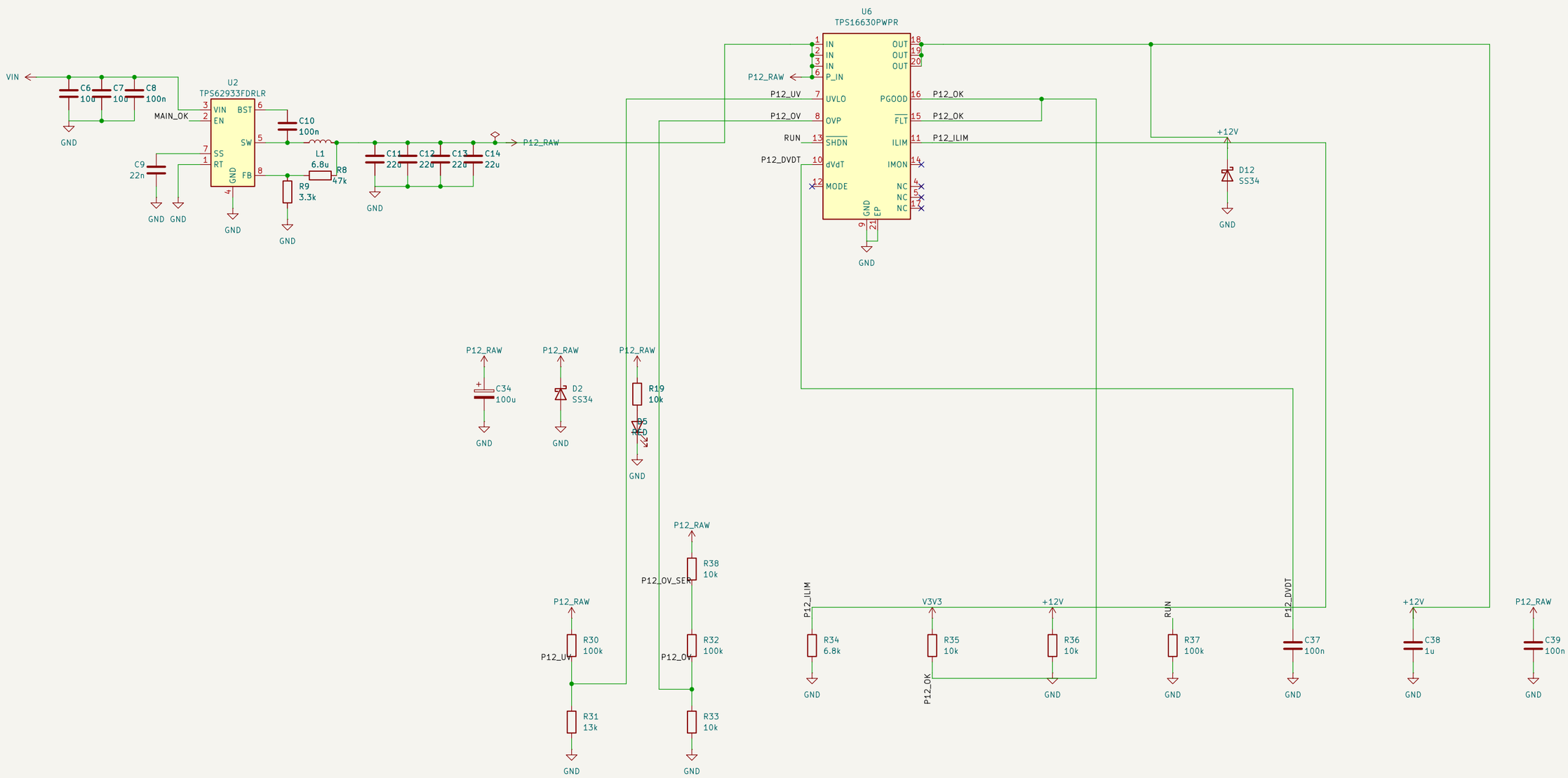
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Sheet: /USB PD and main eFuse/
File: 01_usb_input.kicad_sch

Title: USB PD and main eFuse

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 2/8

+12 V CONVERTER AND OUTPUT EFUSE



TPS62933F converter unchanged. RAW feedback/bulk stay BEFORE U6.
U6: 2.647 A nominal, 2.335–2.971 A screened. No guaranteed 2.5 A minimum; MODE latch-off.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

RUN ◇ RUN

MAIN_OK ◇ MAIN_OK

V3V3 ◇ V3V3

P12_OK ◇ P12_OK

VIN ◇ VIN

P12_RAW ◇ P12_RAW

Schematic only — see design.md and s6-rail-protection.md

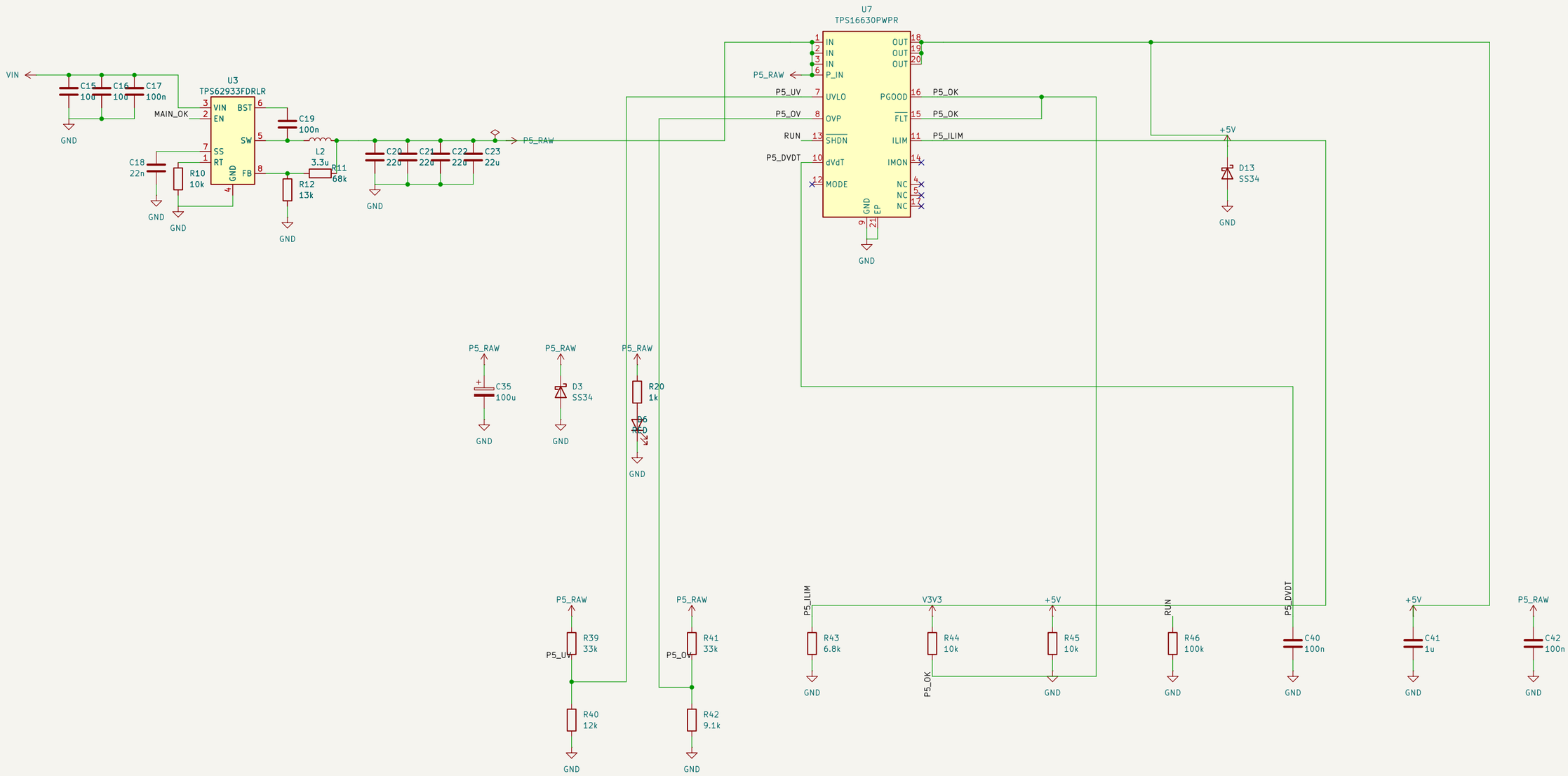
Big Modular

Sheet: /+12 V converter and output eFuse/
File: 02_positive_12v.kicad_sch

Title: +12 V converter and output eFuse

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 3/8

+5 V CONVERTER AND OUTPUT EFUSE



TPS62933F converter unchanged. U7 input must remain >=4.5 V.
U7: 2.647 A nominal, 2.335–2.971 A screened. Output-valid and FLT/PGOOD share P5_OK.

INTER-SHEET INTERFACE (see wired power-flow overview)

+5V ◇ +5V

V3V3 ◇ V3V3

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

P5_OK ◇ P5_OK

RUN ◇ RUN

Schematic only — see design.md and s6-rail-protection.md

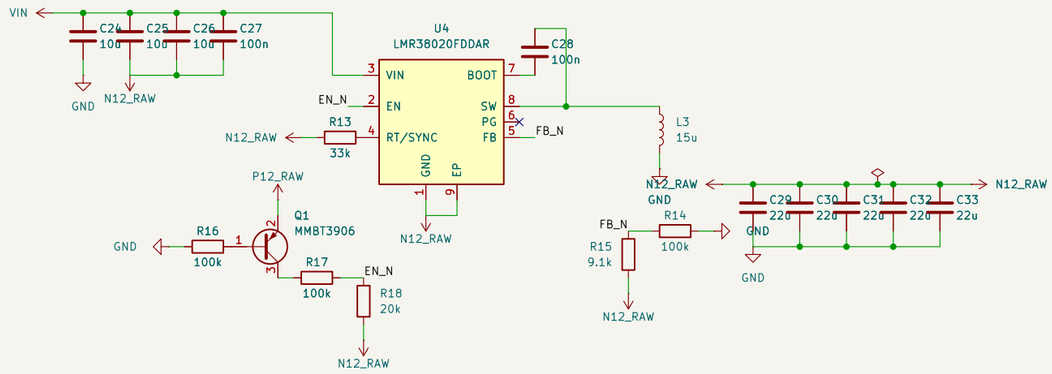
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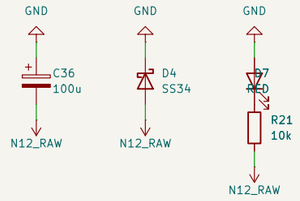
Sheet: /+5 V converter and output eFuse/
File: 03_positive_5v.kicad_sch

Title: +5 V converter and output eFuse

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 4/8

NEGATIVE CONVERTER — RAW RAIL





LMR38020F ground/exposed pad = N12_RAW, not system GND.
Q1 follows P12_RAW: load reaches N12_RAW ONLY through page 05 series protection.

INTER-SHEET INTERFACE (see wired power-flow overview)

N12_RAW ◇ N12_RAW

P12_RAW ◇ P12_RAW

VIN ◇ VIN

Schematic only — see design.md and s6-rail-protection.md
Big Modular

Sheet: /Negative converter — raw rail/
File: 04_negative_converter.kicad_sch

Title: Negative converter — raw rail

Size: User	Date:	**Rev: S6**
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NEGATIVE RAIL PROTECTION AND LEVEL SHIFTING

System GND is continuous. R48 || R49 = 50 mOhm; Q4 is in the NEGATIVE conductor.
U8 VCC is resistor-fed from VIN. U16 disables via UVLO. U17 level shifts PGD.
Nominal 1 A / 3.1 W MOSFET limit. Reverse-collapse pin clamps do not block reverse power.

N_VCC

VIN
R50
10k
N_VCC

VIN
R51
10k
N_VCC

N_VCC
C43
1u
N12_RAW

U8
LM5067MM-2/NOPB
N_VCC 1 VCC
N_UV 2 UVLO
N_OV 3 OVLO
N_PWR 4 PWR
N12_RAW 5 VEE
N_TIMER 6 TIMER
SENSE 7 N_SENSE_PIN
GATE 8 N_GATE
OUT 9 N_OUT_PIN
PGD 10 N_PGD

N_SENSE
R89
33
N_SENSE_PIN
R90
10k
N_OUT_PIN

−12V
Q4
FQD13N06LTM
N_GATE
N_SENSE

N_SENSE
R48
0.1
N12_RAW

N_SENSE
R49
0.1
N12_RAW

V3V3
R59
1k
NEG_OFF_LED

V3V3
R60
1k
NEG_OFF_LED

U16
LTV-354T
NEG_OFF_LED
NEG_OFF_K
N_UV
N12_RAW

NEG_OFF_K
Q5
2N7002
NOT_RUN
GND

GND
R61
10k
N_PGD_LED

V3V3
R62
100k
N12_STATUS

U17
LTV-354T
N_PGD_LED
N_PGD
N12_STATUS
GND

GND
R52
68k
N_UV

GND
R54
47k
N_OV

N_PWR
R56
22k
N12_RAW

N_TIMER
C44
1u
N12_RAW

N_GATE
R57
100k
N_SENSE

R53
33k
N12_RAW

R55
10k
N12_RAW

GND
C45
1u
−12V

GND
R58
10k
−12V

BAT54SLT1G
N12_RAW
N_SENSE_PIN
D19
N_SENSE_PIN

BAT54SLT1G
N12_RAW
N_OUT_PIN
D20
N_OUT_PIN

GND
D14
SS34
−12V

−12V

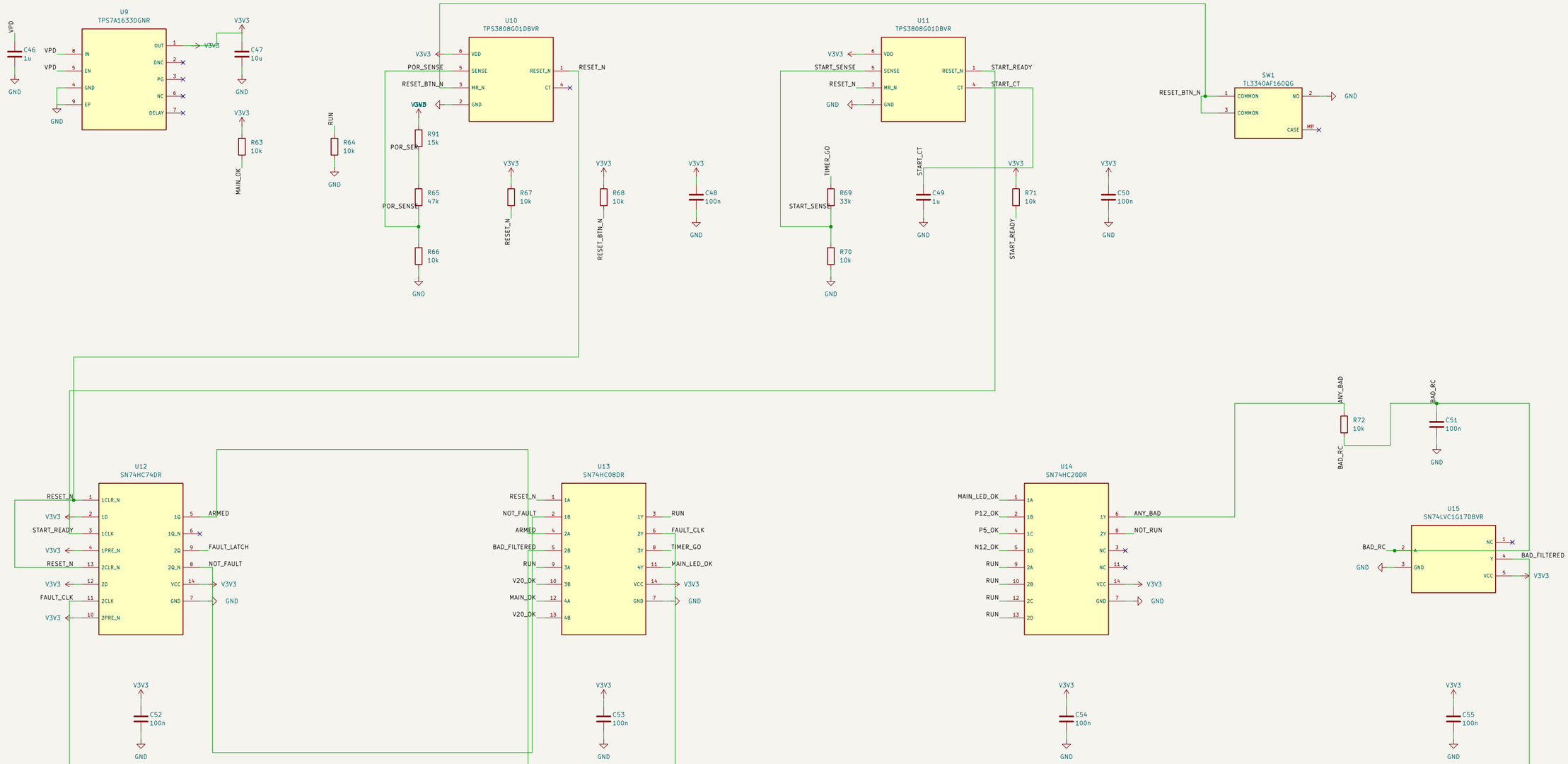
INTER-SHEET INTERFACE (see wired power-flow overview)

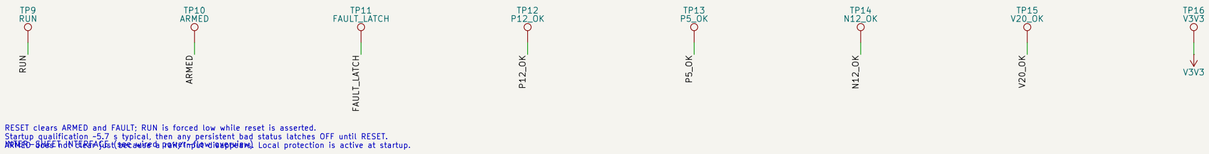
−12V −12V
N12_RAW N12_RAW
N12_STATUS N12_STATUS
NOT_RUN NOT_RUN
V3V3 V3V3
VIN VIN

Schematic only — see design.md and s6-rail-protection.md
Big Modular
Sheet: /Negative rail protection and level shifting/
File: 05_negative_protection.kicad_sch
Title: Negative rail protection and level shifting

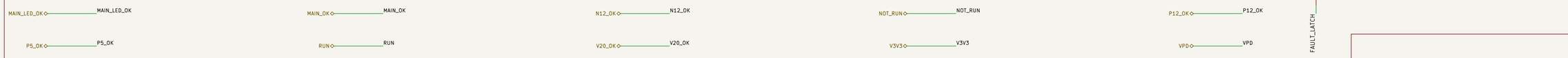
Size: User	Date:	**Rev: S6**
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ALWAYS-ON CONTROL — ALL-RAIL FAULT LATCH AND RESET





RESET clears ARMED and FAULT; RUN is forced low while reset is asserted.
Startup qualification ~5.7 s typical, then any persistent bad status latches OFF until RESET.
ARMED sheet interface (see wires & power inputs elsewhere). Local protection is active at startup.



Schematic only — see design.md and s6-rail-protection.md

Big Modular

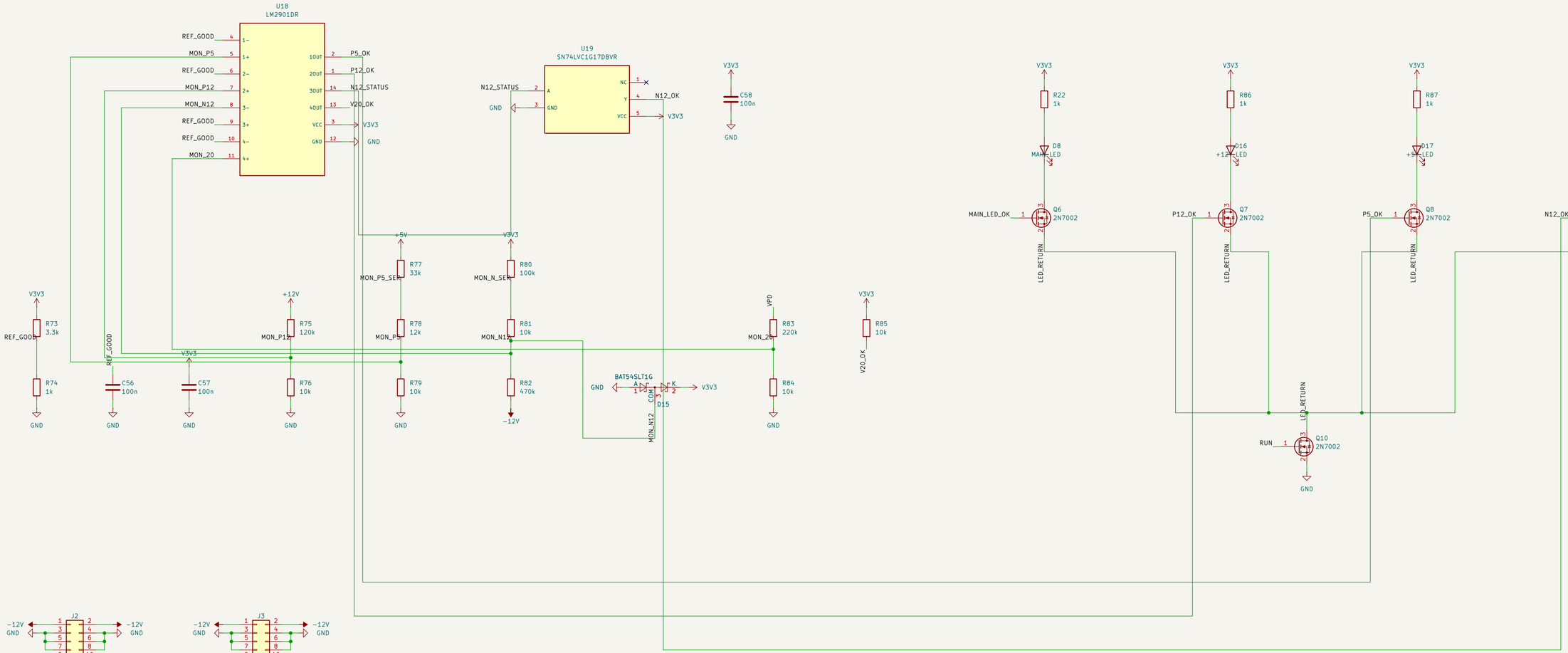
Sheet: /Always-on control — all-rail fault latch and RESET/
File: 06_fault_reset.kicad_sch

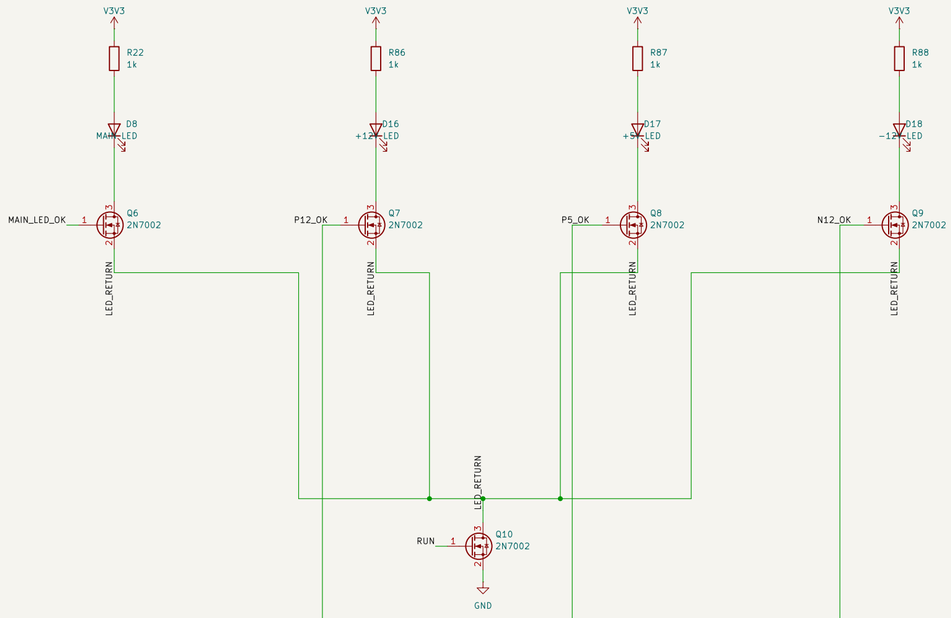
Title: Always-on control — all-rail fault latch and RESET

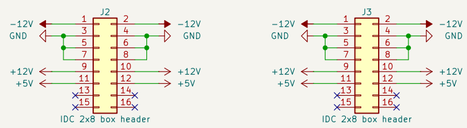
Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 7/8

PROTECTED OUTPUTS — VOLTAGE MONITORS AND PANEL LEDS

D8 MAIN: input-valid AND main-eFuse status AND RUN. D16/+12, D17/+5, D18/−12: delivered-rail status AND RUN.
Any latched common fault turns all four panel LEDs OFF. D5/D6/D7 on converter pages indicate RAW rails.
Voltage-present thresholds are not +/−5% regulation guarantees: retained external charge is not actively discharged.







INTER-SHEET INTERFACE (see wired power-flow overview)

+12V
N12_STATUS
V3V3
+5V
P12_OK
VPD
−12V
P5_OK
MAIN_LED_OK
RUN
N12_OK
V20_OK

Schematic only — see design.md and s6-rail-protection.md

Big Modular

Sheet: /Protected outputs — voltage monitors and panel LEDs/
File: 07_outputs_status.kicad_sch

Title: Protected outputs — voltage monitors and panel LEDs

Size: User	Date:	**Rev: S6**
KiCad E.D.A. 10.0.6		Id: 8/8